Hausdorff approximation of a polytope from random approximate-optimization queries

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Editorial note. Only formal upper-bound-3 and its informal Hausdorff description count as one task. Complete author body/support retained; local inputs and author bibliography mechanically expanded.

Allowed standard prerequisites (editorial).

- Standard algorithmic and mathematical background cited by the authors. All source-local constructions, proofs, appendices and references are retained; no new mathematical assumptions added.

Abstract

The Separating Hyperplane theorem is a fundamental result in Convex Geometry with myriad applications. The theorem asserts that for a point a not in a closed convex set K, there is a hyperplane with K on one side and a strictly on the other side. Our first result, Random Separating Hyperplane Theorem (RSH), is a strengthening of this for polytopes. RSH asserts that if the distance between a and a polytope K with k vertices and unit diameter in $\Re^d$ is at least δ, where δ is a fixed constant in $(0, 1)$, then a randomly chosen hyperplane separates a and K with probability at least $1/\operatorname{poly}(k)$ and margin at least $\Omega\left(\delta/\sqrt{d}\right)$. There is a rich body of work on reductions between (approximate) optimization and (approximate) separation oracles for general convex sets, where the focus has been on the number of oracle calls. An immediate consequence of our result is the first near optimal bound on the error increase in the reduction from a Separation oracle to an Optimization oracle over a polytope.

RSH has algorithmic applications in learning polytopes. We consider a fundamental problem, denoted the "Hausdorff problem", of learning a unit diameter polytope K within Hausdorff distance δ, given an optimization oracle for K. Using RSH, we show that with polynomially many random queries to the optimization oracle, K can be approximated within error $O(\delta)$. To our knowledge this is the first provable algorithm for the Hausdorff Problem. Building on this result, we show that if the vertices of K are well-separated, then an optimization oracle can be used to generate a list of points, each within Hausdorff distance $O(\delta)$ of K, with the property that the list contains a point close to each vertex of K. Further, we show how to prune this list to generate a (unique) approximation to each vertex of the polytope. We prove that in many latent variable settings, e.g., topic modeling, LDA, optimization oracles do exist provided we project to a suitable SVD subspace. Thus, our work yields the first efficient algorithm for finding approximations to the vertices of the latent polytope under the well-separatedness assumption. This assumption states that each vertex of K is far from the convex hull of the remaining vertices of K, and is much weaker than other assumptions behind algorithms in the literature which find vertices of the latent polytope.

1 Introduction

The Separating Hyperplane theorem (SHT) is a fundamental result in Convex Geometry with myriad applications (see e.g. [Ber09]). The theorem asserts that for a point a not in a closed convex set K, there is a hyperplane with K on one side and a strictly on the other side.

This paper makes two main contributions. Our theoretical contribution, which is an extension of the classical SHT, is what we call the Random Separating Hyperplane Theorem (RSH). Our main algorithmic contribution is to use RSH to prove that a natural algorithm, which we call the k-OLP algorithm, can learn (vertices of) latent polytopes arising in a number of problems in Latent Variable Models including Clustering, Mixture Learning , LDA (linear discriminant analysis), Topic Models. The algorithmic result is shown by reducing the problem of learning latent polytopes in a variety of settings to that of constructing approximate optimization oracles for the corresponding polytopes. Bulk of our algorithmic contribution is in proving this reduction and the existence of such oracles.

1.1 Random Separating Hyperplane Theorem(RSH)

RSH draws its motivation mainly from the Separating Hyperplane Theorem (SHT) of Convex Geometry. It also has connections to the Johnson-Lindenstrauss Random Projection theorem [JL84] and reductions among oracles in Convex Optimization. The SHT formally states that given a closed convex set K and a point $a \notin K$, there exists a (non-zero) vector u such that

$$u \cdot a \; > \; \mathrm{Max}_{y \in K}\, u \cdot y.$$

The following question arises: Does a randomly picked u separate a from K? Taking into account some necessary conditions for a positive answer, we can ask if the following inequality holds for a randomly chosen u: (here Δ is the diameter of K, a is at distance at least $\delta\Delta$ from K, where $\delta \in (0,1)$):

$$\Pr\left(u \cdot a \geq \mathrm{Max}_{y \in K} u \cdot y + |u|\alpha\delta\Delta\right) \; \geq \; 1/\mathrm{poly}_{\delta},^{1} \tag{1}$$

with α being as high as possible.

The question (1) is also motivated from the Johnson-Lindenstrauss Random Projection theorem (JL theorem) [JL84] which states that if a, b are points in R^d and U is a random subspace of dimension s, then with probability bounded away from 0, the distance between the projection of a and b on U is at least $\Omega(|a-b|\sqrt{s}/\sqrt{d})$. The following natural generalization of this is interesting already for $s = 1$:

Instead of b being a point, if it is now a polytope K, does a similar lower bound on the distance of a to K in the projection onto a random line hold?

It is easy to see that in spirit, this is the same question as whether (1) holds. It is also easy (see below) to see that the projection shrinks the distance between a and K by a factor of $\Omega^*(\sqrt{d})$. The RSH theorem proves that the shrinkage is $O(\sqrt{d})$ thus making this parameter nearly (within log factors) tight. We now state the RSH theorem:

Theorem 1.1 (Random Separating Hyperplane Theorem(RSH): Informal version). Suppose K is a k vertex polytope with diameter Δ and a is a point at distance at least $\delta\Delta$, $\delta \in (0,1)$, from K. Let V be an $m-$dimensional subspace containing $K \cup \{a\}$. For a random Gaussian vector $u \in V$, the following event happens with probability at least $1/\mathrm{poly}_{\delta}(k)$:

$$u \cdot a \geq \mathrm{Max}_{y \in K} u \cdot y + \frac{\delta\Delta|u|}{10\sqrt{m}}. \tag{2}$$

We provide a simple example where K is a line segment (see Section) to show that the factor $\sqrt{m}$ cannot be improved. It is also interesting to note that the success probability of the event in (2)

[1] $\mathrm{poly}_{\delta}(z)$ denotes $z^{\mathrm{poly}(1/\delta)}$.

needs to depend on k. (see Section). In particular, RSH does not hold for general convex sets (where k is not necessarily finite.)

1.2 Oracles for Convex Sets

The seminal work of [GLS88] showed that the Ellipsoid Algorithm can be viewed as a reduction of an Optimization Oracle to a Separation Oracle for convex sets. Since then, there has been an extensive study of reductions among approximate oracles (also referred to as "weak" oracles). There are two important parameters in a reduction from an oracle $\mathcal{A}$ with error δ to an oracle $\mathcal{B}$ with error ε – the number of calls to $\mathcal{B}$ used and the increase in error, namely, δ/ε. Of these, the number of oracle calls has received much attention since it is a measure of running time. But the error parameter has also been taken into account in most results starting with [GLS88]. The best known bounds on the number of oracle calls in reductions are due to [LSV18]; they achieve near-linear number of oracle calls. The known error increase factors are $\Omega(d)$, where d is the dimension. Our proof of RSH gives a simple polynomial time reduction (for fixed δ) from Separation to Optimization with the error increase factor of $O^*(\sqrt{d})$ which we also show is best possible within log factors.

We define our approximate oracles (which differ from the traditional definitions of approximate oracles – see Remark 1.4) and state our nearly-matching upper and lower bounds for error increase in the reduction. Here, B_d denotes the unit ball in $\Re^d$.

Definition 1.2. For a non-empty convex set $K \subseteq \Re^d$ and $\delta \in (0,1)$, a separation oracle for K with error δ, denoted $\mathsf{SepOr}_\delta(K)$ oracle, takes as input any $a \in \Re^d$ and returns a valid option among the two below (Note: Both may be valid):

- $a \in K + \delta\Delta(K)B_d$.
- Returns u satisfying $u \cdot a > \text{Max}_{y \in K} u \cdot y$

Definition 1.3. For a non-empty convex set $K \subseteq \Re^d$ and $\varepsilon \in (0,1)$, an Optimization oracle for K with error ε, denoted $\mathsf{OptOr}_\varepsilon(K)$ oracle, takes as input any $u \in \Re^d, |u| = 1$, and returns a point $x(u)$ satisfying both these conditions :

- $x(u) \in K + \varepsilon\Delta(K)B_d$, and
- $u \cdot x(u) \geq \text{Max}_{y \in K} u \cdot y - \varepsilon\Delta(K)$

Remark 1.4. Starting with [GLS88], the second option in the traditional definition of approximate oracles usually replaces the K we have in that option with a subset of K, namely, the subset of all points with the property that a ball of specified size centered at that point is wholly inside K. This is necessary since the proof of convergence of the Ellipsoid algorithm is by shrinking the volume of the ellipsoid containing K. If K is not full dimensional, the subset of K is empty and a worst-case (adversarial) oracle can always return this option giving us no information on K. Our definition makes the stronger assumption with K in the second option, thus, dealing with (among other examples) non-full dimensional K.

Theorem 1.5. Let $\delta \in (0,1)$ be any constant and K be a polytope in $\Re^d$ with $k \geq 1$ vertices given by an $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{A}$, with $\varepsilon \leq \delta/100\sqrt{d}$. Then there is an $\mathsf{SepOr}_\delta(K)$ oracle which is obtained by making $\text{poly}_\delta(dk)$ calls to $\mathcal{A}$.

It is worth noting that the calls to optimization oracle in the above result use random vectors u, and so the reduction algorithm is randomized. The following result shows that the condition $\varepsilon \leq \delta/100\sqrt{d}$ above is almost tight in the sense no deterministic algorithm can beat it by more than log factors.

Theorem 1.6. There is no deterministic polynomial time reduction from an $\mathsf{SepOr}_{O(1)}(K)$ oracle to an oracle in $\mathsf{OptOr}_{\Omega(\log d/\sqrt{d})}(K)$.

1.3 Algorithmic application of RSH

We now discuss the second main contribution of our work, i.e., applications of RSH to learning vertices of a latent polytope.

Problem Formulation. Several latent variable problems including Clustering, LDA, MMBM can be reduced (See Section 7 for details) to a problem that we call k-OLP: given an ε-optimization oracle for a k vertex polytope $K \subseteq \mathrm{R}^d$, learn the vertices of K (approximately). We define two simpler (than k-OLP) problems – ListLearn and Hausdorff, that are related to k-OLP, and then we define k-OLP.

The first problem, Hausdorff, seeks to find approximation to a polytope K when we are given an approximate optimization oracle for the polytope.

Definition 1.7 ((ε,δ)-Hausdorff-Problem). Given an $\mathsf{OptOr}_\varepsilon(K)$ oracle for a polytope K in R^d with k vertices, find a set P of $m = \mathrm{poly}_\delta(dk)$ points such that $\mathsf{Haus}(CH(P), K) \leq \delta\Delta(K)$, where, Haus denotes Hausdorff distance (see Definition 5.1 for a formal definition).

In the problem ListLearn, we also wish to find a small list of points, such that each vertex of K is close to at least one point in this list.

Definition 1.8. [(ε,δ)-ListLearn Problem] Given an $\mathsf{OptOr}_\varepsilon(K)$ oracle for a polytope K in $\Re^d$ with k vertices, each separated from the convex hull of the other $k-1$ vertices by at least $\delta\Delta(K)$, find a list $P \subseteq K + \delta\Delta(K)B_d$ of $m = \mathrm{poly}_\delta(dk)$ points such that for every vertex v of K, there is some $v' \in P$ with $|v - v'| \leq \delta\Delta(K)/10$.

When the parameters ε, δ will be clear from the context, we shall abbreviate the above two problems as Hausdorff and ListLearn problems respectively. It is not difficult to see that any solution P to ListLearn is also a solution to Hausdorff, but, the converse need not hold: $\mathrm{CH}(P)$ may nearly contain K without P having any point close to some vertex of K. Our technical results (see below for the informal versions) show that if $\varepsilon \in O_\delta(1/\sqrt{d})$ [2], then we can solve the above-mentioned problems efficiently and indeed then, the following simple algorithm gives the desired answers (the proof crucially uses RSH):

> Random Probes Algorithm
>
> - Pick uniformly at random unit vectors $u_1, u_2, \ldots u_m$, where $m = \mathrm{poly}_\delta(dk)$.
> - Return P, which is the set of m answers of the $\mathsf{OptOr}_\varepsilon(K)$ oracle to the queries $u_1, u_2, \ldots u_m$.

The first result (see Theorem 5.2 for a formal statement) states that the convex hull of answers to polynomially many random queries to the approximate optimization oracle approximates K well.

Theorem 1.9 (Hausdorff Approximation from oracle (Informal)). Consider an instance of the (ε,δ)-Hausdorff problem for a polytope $K \subseteq \Re^d$. Assume that $\varepsilon \in O_\delta(1/\sqrt{d})$, and let P be the set of points returned by the Random Probes Algorithm above. Then with high probability,

$$\mathsf{Haus}(CH(P), K) \leq \delta\Delta(K).$$

[2] $O_\delta(x)$ stands for $f(\delta)x$ for some function f.

The second result (see Theorem 6.5 for a formal statement) shows that as long as each vertex of K is well-separated from the convex hull of the other vertices of K, the set P constructed by the Random Probes Algorithm contains an approximation to each of the vertices. Thus, answers to polynomially many random queries list-learns the polytope.

Theorem 1.10 (List-Learning from Oracle (Informal)). Consider an instance of the ListLearn problem for a polytope $K \subseteq \Re^d$, and assume that $\varepsilon \in O_\delta(1/\sqrt{d})$. Then, with high probability, the set P output by the Random Probes Algorithm above has the following property: for every vertex a of K, there is a point $a' \in P$ with

$$|a' - a| \leq \delta\Delta(K)/10.$$

The $\sqrt{d}$ factor in both the above theorems is near-optimal (within a $\log d$ factor). Indeed, we shall prove:

Theorem 1.11 (Oracle Lower Bound). The problem where, one is required to output a point which is within $\Delta(K)/10$ of some vertex of K, given only by an $\mathsf{OptOr}_\varepsilon(K)$ oracle, cannot be solved in deterministic polynomial time when $\varepsilon \geq 8 \ln d/\sqrt{d}$.

We now define the k-OLP problem (the parameters ε, δ in the definition will often be clear from the context and may not be mentioned explicitly). The problem statement is similar to that of ListLearn, but we want to output a list of exactly k points.

Definition 1.12. [(ε, δ)-k-OLP Problem] Under the same hypothesis as for the ListLearn problem, find a set of points P, $|P| = k$, satisfying the following condition: for each vertex v of K, there is a (unique) point $v' \in P$ such that $|v - v'| \leq \delta\Delta(K)/10$.

Our next result gives a strengthening of Theorem 1.10.

Theorem 1.13. Consider an instance of the k-OLP problem on a polytope $K \subseteq \Re^d$, and assume $\varepsilon \in O_\delta(1/\sqrt{d})$ in a k-OLP problem. Let P be the set of points returned by the Random Probes Algorithm. Then, in polynomial time, we an find a $Q \subseteq P, |Q| = k$ satisfying the following condition: for each vertex v of K, there is a (unique) $v' \in P, |v - v'| \leq \delta\Delta(K)/10$.

The algorithm for finding Q from P is likely of independent interest. We call this problem the "Soft Convex Hull" problem and it is described in Section 7.1.

Do Approximate Optimization Oracles exist? The answer to this question is a qualified Yes. They exist, but unfortunately, as we point out below, for many latent variable problems including the simple mixture of two Gaussians with means separated by $\Omega(1)$ standard deviations, we do not get $\varepsilon \in O_\delta(1/\sqrt{d})$, when, $k < d$. Thus, we do not satisfy the hypothesis of the results mentioned in Theorem 1.9,Theorem 1.10, and Theorem 1.13. But we are able to tackle this hurdle by projecting to the k-SVD subspace (of the input data points which satisfy conditions discussed below) where, we do get the necessary $\varepsilon \in O_\delta(1/\sqrt{k})$.

First we observe that approximate optimization oracles arise in a natural setting – that of latent variable models. [BK20] show that these models can be reduced to a geometric problem called LkP described below. [We will not reproduce the reduction here.] LkP is the following problem: Let K be a k vertex polytope in $\Re^d$. Let $M_{\cdot,1}, \ldots, M_{\cdot,k}$ denote the vertices of K. Assume that there are latent (hidden) points $P_{\cdot,j}, j = 1, 2, \ldots, n$, in K. The observed data points $A_{\cdot,j}, j = 1, 2, \ldots, n$ are generated (not necessarily under any stochastic assumptions) by adding displacements $A_{\cdot,j} - P_{\cdot,j}$ respectively to $P_{\cdot,j}$. Let[3]

$$\sigma_0 := \frac{||\mathrm{P} - \mathrm{A}||}{\sqrt{n}}.$$

[3]By the standard definition of spectral norm, it is easy to see that σ_0^2 is the maximum mean squared displacement in any direction.

We assume that there is a certain w_0 fraction of latent points close to every vertex of K, i.e., for all $\ell \in [k]$,

$$C_\ell := \{j : |P_{\cdot,j} - M_{\cdot,\ell}| \leq \frac{\sigma_0}{\sqrt{w_0}}\} \text{ satisfies } |C_\ell| \geq w_0 n.$$

Theorem 1.14 (From Data to Oracles). Using the above notation, the following "Subset Smoothing algorithm" gives us a polynomial time $\mathsf{OptOr}_{\frac{4\sigma_0}{\Delta\sqrt{w_0}}}(K)$ oracle..

> Subset Smoothing Algorithm
> Given query u, let S be the set of the $w_0 n$ j's with the highest $u \cdot A_{\cdot,j}$ values.
> Return $A_{\cdot,S} := \frac{1}{w_0 n} \sum_{j \in S} A_{\cdot,j}$.

The Subset Smoothing algorithm was used in [BK20]. It is also reminiscent of Superquantiles [RR14], though our use here is not directly related to them. While this theorem helps us get optimization oracles, the error guarantee of $O(\sigma_0/\Delta\sqrt{w_0})$ is not good enough in many applications. An elementary example illustrates this issue:

Consider a mixture of two equal weight standard Gaussians centered at $-v$ and v, where, v is a vector of length 10. [This fits the paradigm "means separated by $\Omega(1)$ standard deviations".] Then, data generated by the mixture model fits our data generation process with $K = \{\lambda v, \lambda \in [-1, 1]\}$, and each $P_{\cdot,j}$ is either v or $-v$ depending on the Gaussian from which the point has been sampled. Here $A_{\cdot,j}$ denotes the actual sampled point from the mixture. Now, $\Delta = 20$ and it can be seen from Random Matrix Theorems (see e.g., [Ver10]) that $\sigma_0 = O(1)$ with high probability. So, $\sigma_0/\Delta\sqrt{w_0} \in O(1)$ with high probability, and hence, the Theorem above yields an $\mathsf{OptOr}_\varepsilon(K)$ oracle with $\varepsilon \in \Omega(1)$. But d can be arbitrarily large and so we do not have the required $\mathsf{OptOr}_{O(1/\sqrt{d})}(K)$ oracle.

This elementary example can be tackled in several ways. Our algorithm which we call the "k-OLP algorithm" is simply stated and works in general settings (including on this toy example) for several Latent Variable problems (see Section 7 for details). The main idea is to first project the input points on a suitable SVD subspace and then use the approximate optimization oracle in the projection.

SVD and the k-OLP Algorithm We now state the result (see Theorem 8.1 for a formal version) for k-OLP in the setting of LkP. As mentioned above, this uses SVD followed by subset smoothing.

Theorem 1.15. Recall the notation and assumptions of Theorem 1.14. In addition, we assume that each vertex of K is $\delta\Delta(K)$ far from the convex hull of other vertices of K, where, δ satisfies:

$$\sigma_0 \leq c\delta^2\Delta\sqrt{w_0}/\sqrt{k}.$$

Then, the set of points P returned by the following k-OLP algorithm list learns the vertices of K. Further, we can find a subset Q of P with $|Q| = k$ and for each v, vertex of K, Q contains a v' with $|v - v'| \leq \delta\Delta/10$:

> Algorithm k-OLP
> 1. Project to the k-dim SVD subspace V corresponding to the points $A_{\cdot,j}$.
> 2. Pick $m = \text{poly}_\delta(k)$ random vectors $u_1, u_2, \ldots, u_m$ in V.
> 3. For each u_i, take the mean of the $A_{\cdot,j}$ with the $w_0 n/2$ highest values of $u_i \cdot A_{\cdot,j}$.
> 4. Let P be the set of m means computed in the step above.
> 5. Output a subset Q of P, $|Q| = k$, using Theorem 1.13.

We sketch here the steps in the proof (the details are contained in the proof of Theorem 8.1.)

Sketch. Let $\widehat{K}$ denote projection of K onto V. By Theorem 1.14 , Step 3 of Algorithm k-OLP is an $\mathsf{OptOr}_{\varepsilon}(\widehat{K})$ oracle, where $\varepsilon = O\left(\frac{\sigma_0}{\Delta\sqrt{w_0}}\right)$. Also, each $\widehat{M}_{\cdot,\ell}$, which is the projection of $\widehat{M}_{\cdot,\ell}$ on V, is $O(\delta\Delta(K))$ far from the convex hull of the other vertices of $\widehat{K}$ (See Lemma 8.3.) Now, Theorem 1.13 applied to $\widehat{K}$ in the subspace V implies the desired result. □

It is worth noting that data obtained from several generative models are known to satisfy the LkP condition stated in Theorem 1.15, e.g., Stochastic Mixture models with k components, Topic Models, Mixed membership community models.

From List Learning to k-OLP As outlined above, the k-OLP algorithm works in two stages: (i) Project the data points on the SVD subspace V of dimension k, and (ii) make polynomially calls to the $\mathsf{OptOr}_{\varepsilon}(K)$ oracle, where each query is given by a randomly chosen unit vector in the subspace V (as in the statement of Theorem 1.9) – let P be the set of points returned by the oracle. The first statement in Theorem 1.15 shows that the convex hull of P is close to K.

Obtaining approximations to the vertices of K from P requires addressing a new problem: given a set of points W, find a small subset T of W, such that their convex hulls are close. We call this the soft convex hull problem. A similar problem was addressed by [BHR19]; however they gave a bi-criteria approximation algorithm for this problem. Under stronger assumptions, where we assume that there in the optimal solution $T^\star$, each point of $T^\star$ is well-separated from the convex hull of the rest of the points of $T^\star$, we show that one can recover approximations to each of the points in $T^\star$. Applying this result to the set of points P returned by the optimization oracle, we get a set of k points Q, each of which approximates a unique vertex of the polytope K.

The algorithm for obtaining soft convex hull proceeds as follows. We first prune points $w \in W$ which have the following property: consider the subset X of points in W which are sufficiently far from w. Then w is close the convex hull of X. After pruning such points from W, we pick a subset of points which are sufficiently far-apart from each other. The main technical result shows that this procedure outputs the desired set T.

1.4 Related Work

The seminal work of [GLS88] showed that optimization of a convex function over a convex set can be reduced to separation oracles using the classical ellipsoid algorithm. There has been active research on reducing the number of separation (or membership) oracle queries (see e.g. [AH15, KV06, LV06]). [LSV18] showed that, for a "well-rounded" convex set K (i.e., K has unit diameter and contains a ball of constant radius), we can get a separation oracle by making $\widetilde{O}(d)$ calls to an optimization oracle for K. In this reduction, in order to get a separation oracle with error δ, they use an optimization oracle with error $\mathrm{poly}(\delta/d)$. As mentioned in the previous section, RSH Theorem implies that we can obtain a separation oracle with error δ for a convex polytope K from an optimization oracle with error $O(\delta/\sqrt{d})$.

The well known result of [JL84] shows that given a set of n points in $\Re^d$, projection to a random subspace of dimension $O(\log n/\varepsilon^2)$ preserves all pair-wise distances up to $(1+\varepsilon)$-factor with high probability. Further, this bound on the dimension on which the points are projected is known to be tight [Alo03, LN17]. Note that in our setting, there are $k+1$ points "of interest", namely, the k vertices of K and a point $a \notin K$ and by the above, a random projection to $O^*(\log k)$ dimensional space preserves all pairwise distances among them. But this is not sufficient for our problems. We need separation of a from all of K in the projection. We achieve this by projecting to a set of random 1-dimensional subspaces, and show that the distance between a point and a polytope does not scale

down by more than $O(\sqrt{d})$ factor for at least one of them with high probability (this is an immediate corollary of the RSH Theorem).

The problem of learning vertices of a polytope arises in many settings where data is assumed to be generated by a stochastic process parameterized by a model. Examples include topic models [Ble12], stochastic block models [ABEF14], latent Dirichlet allocation [BNJ03]. A variety of techniques have been developed for these specific problems (see e.g. [AFH+15, AGH+18, HS17]). [BK20] (see also [BBK+21]) proposed the latent k-polytope (LkP) model which seeks to unify all of these latent variable models. In this model, there is latent polytope with k vertices, and data is generated in a two step process: first we pick latent points from this polytope, and then the observed points are obtained by perturbing these latent points in an adversarial manner. They showed that under suitable assumptions on this deterministic setting, one can capture the above-mentioned latent variable problems. Assuming strong separability conditions on the vertices of the polytope (i.e., each vertex of K is far from the affine hull of other vertices of K), they showed that one can efficiently recover good approximations to the vertices of the polytope from the input data points. In comparison, our assumption on K is that each vertex of K is far from the convex hull of the remaining vertices of K. This is a much milder condition, e.g., it allows a polytope with more than 2 vertices in a plane. [BKK21] showed how to infer the parameter k from data in the LkP setting (under the strong separation condition).

[BHR19] addressed a problem similar to the Hausdorff problem: instead of an ε-optimization oracle for a polytope K, we are given an explicit set P of points, whose convex hull is within Hausdorff distance at most $\delta\Delta(K)$ from K. They are able to get better dependencies on the parameters k, δ, ε than Theorem 5.2 under these stronger assumptions.

2 Preliminaries

For two points $x, y \in \Re^d$, $|x - y|$ denotes the Euclidean distance between the points. Given a point $x \in \Re^d$ and a subset $X \subseteq \Re^d$, define $\mathsf{dist}(x, X)$ as the minimum distance between x and a point in X, i.e., $\inf_{y \in X} |x - y|$. For a set of points X, $\Delta(X)$ denotes the diameter of X, i.e., $\sup_{x,y \in X} |x - y|$. We denote the convex hull of X by $\mathsf{CH}(X)$. For two subsets A, B of $\Re^d$, define their Minkowski sum $A + B$ as $\{x + y : x \in A, y \in B\}$. Similarly, define λA, where $\lambda \in \Re$, as $\{\lambda x : x \in A\}$. For an $m \times n$ matrix B, we use $B_{\cdot,j}$ to denote the jth column of B. For a subset $S \subseteq [n]$ of columns of B, $B_{\cdot,S}$ denotes $\frac{1}{|S|} \sum_{j \in S} B_{\cdot,j}$. Often, we represent the vertices of a polytope K in $\Re^d$ by a $d \times k$ matrix M, and so the columns $M_{\cdot,1}, \ldots, M_{\cdot,k}$ would represent the vertices of K. We shall use the notation $\mathrm{poly}_\delta(z)$ to denote a quantity which is $z^{\mathrm{poly}(1/\delta)}$. Further the notation $O_\delta(z)$ shall denote a quantity which is $f(\delta)z$, where $f(\delta)$ is a function depending on δ only (and hence, is constant if δ is constant).

We now give an outline of rest of the paper. In Section 4, we prove the Random Separating Hyperplane theorem. In Section 5, we prove Theorem 1.9 by showing that an $\mathsf{OptOr}_\varepsilon(K)$ oracle leads to efficient constructions of approximation to K. In Section 6 we give an algorithm for the ListLearn problem under the stronger assumption that the vertices of K are well separated. We also prove the lower bound result Theorem 1.11 in this section. In Section 7, we extend the algorithm for ListLearn to the k-OLP problem. This requires the concept of soft convex hulls. The algorithm for constructing soft convex hulls in given in Section 7.1. Finally, in Section 8, we apply the k-OLP algorithm for the latent polytope problem. As note earlier, in the setting of latent polytopes K, we can only guarantee $\mathsf{OptOr}_\varepsilon(K)$ oracles with ε being $O(1/\sqrt{k})$, whereas our algorithm for k-OLP requires ε to be $O(1/\sqrt{d})$. We handle this issue by projecting to a suitable SVD subspace and executing the k-OLP algorithm in this subspace. We conclude with some open problems in Section 9.

3 Reduction from Separation to Optimization Oracles

In this section, we prove Theorem 1.5, which is reproduced here.

Theorem 1.5. Let $\delta \in (0,1)$ be any constant and K be a polytope in $\Re^d$ with $k \geq 1$ vertices given by an $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{A}$, with $\varepsilon \leq \delta/100\sqrt{d}$. Then there is an $\mathsf{SepOr}_\delta(K)$ oracle which is obtained by making $\text{poly}_\delta(dk)$ calls to $\mathcal{A}$.

Proof. Consider a point $a \in \Re^d$. Use the oracle $\mathcal{A}$ on $\text{poly}_\delta(kd)$ random vectors $u, |u| = 1$. Let U denote the set of these unit vectors. If $a \notin K + \delta\Delta(K)B_d$, then, by RSH, with high probability, there is an $u \in U$ such that $u \cdot a > \text{Max}_{y \in K} u \cdot y + \delta\Delta(K)/(10\sqrt{d})$. For this u, we have (using $x(u) \in K + (\delta\Delta(K)/100\sqrt{d})B_d$):

$$u \cdot a > u \cdot x(u) + (\delta\Delta(K)/11\sqrt{d}). \tag{3}$$

Conversely, for any $u \in U$ satisfying (3), we have (using $u \cdot x(u) \geq \text{Max}_{y \in K} u \cdot y - \delta\Delta(K)/100\sqrt{d}$): $u \cdot a > \text{Max}_{y \in K} u \cdot y + \delta\Delta(K)/20\sqrt{d}$. Thus, we obtain a $\mathsf{SepOr}_\delta(K)$ oracle as follows: check if (3) holds for any $u \in U$. We are guaranteed to find one with high probability if $a \notin K + \delta\Delta B$ and that provides the required separating hyperplane. □

The proof of the lower bound result Theorem 1.6 is very similar to that of Theorem 1.11, which is given in Section 6.

4 Random Separating Hyperplane (RSH) Theorem

In this section, we prove RSH for polytopes: If a point p point is at a distance from a polytope K, then, the Gaussian measure of the set of well-separating hyperplanes has a positive lower bound depending on the number of vertices in K. More specifically, we show:

Theorem 4.1. Suppose K is a polytope in $\Re^d$ with k vertices and diameter $\Delta(K)$. Suppose a is a point in R^d and $\delta \in (0,1]$ with

$$\min_{y \in K} |a - y| \geq \delta\Delta(K). \tag{4}$$

Let V be an m-dimensional subspace containing $\mathsf{Span}(K \cup \{a\})$ and let u be a random vector drawn from the normal distribution $N(0, I_m)$ in V. Then,

$$\Pr_u \left[(u \cdot a - \max_{y \in K} u \cdot y) \geq |u| \cdot \delta\Delta(K) \cdot \frac{\sqrt{\log k}}{\sqrt{\log k} + 4\delta\sqrt{m}} \right] \geq \frac{1}{40} k^{-10/\delta^2}.$$

Proof. Let $\zeta_1, \zeta_2, \ldots, \zeta_k$ be the vertices of K. Let b be the closest point in K to a, and define

$$w = \frac{a - b}{|a - b|}.$$

Then by standard Convex Geoemtry arguments, we have for all points $y \in K$:

$$w \cdot y \leq w \cdot b. \tag{5}$$

Let u be as in the statement of the theorem. We can write u as

$$u = \lambda w + z, \text{ where } z \perp w.$$

In order to prove the theorem, we first express $(u \cdot a - \max_{y \in K} u \cdot y)$ as

$$u \cdot (a-b) - \max_{y \in K} u \cdot (y-b) = u \cdot (a-b) - \max_{\ell=1,\ldots,k} u \cdot (\zeta_\ell - b) \tag{6}$$

It suffices to show that with reasonably high probability, there is a lower bound on $u \cdot (a-b)$ and an upper bound on $u \cdot (\zeta_\ell - b)$ for all ℓ. Observe that

$$u \cdot (\zeta_\ell - b) = \lambda w \cdot (\zeta_\ell - b) + z \cdot (\zeta_\ell - b) \overset{(5)}{\leq} z \cdot (\zeta_\ell - b). \tag{7}$$

Therefore, it suffices to upper bound $z \cdot (\zeta_\ell - b)$ for all vertices ζ_ℓ. We now define a sequence of events and bound the probability of each of these events.

Fact 4.2. Let $\mathcal{E}_0$ be the event $[|z| \leq 4\sqrt{m}]$. Then $\Pr[\neg \mathcal{E}_0] \leq 1/20$.

Proof. The coordinates of z are independent $N(0,1)$ random variables, and since z lies in an $(m-1)$-dimensional subspace, the result follows from standard concentration bounds for sum of squares of Gaussian random variables. □

Fact 4.3. For $\ell = 1, 2, \ldots, k$, define the event $\mathcal{E}_\ell$ as $[|z \cdot (\zeta_\ell - b)| \leq 2\sqrt{\ln k}\Delta(K)]$. Then $\Pr[\neg \mathcal{E}_\ell] \leq \frac{1}{4k^2}$.

Proof. Observe that $z \cdot (\zeta_\ell - b)$ is a 1-dimensional Gaussian random variable with zero mean and with variance $|\zeta_\ell - b|^2 \leq \Delta(K)^2$. The result now follows the tail bounds for a 1-dimensional Gaussian random variable. □

Fact 4.4. Let $\mathcal{E}_{k+1}$ denote the event that $\lambda \geq 3\sqrt{\ln k}/\delta$. Then $\Pr[\neg \mathcal{E}_{k+1}] \geq \frac{1}{10} k^{-10/\delta^2}$.

Proof. Since λ is a 1-dimensional Gaussian random variable, the result follows from tail bounds for normal distribution. □

Corollary 4.5. $\Pr\left[\wedge_{\ell=0}^{k+1} \mathcal{E}_\ell\right] \geq \frac{1}{40} k^{-10/\delta^2}$.

Proof. Applying union bound to Fact 4.2 and Fact 4.3, we get

$$\Pr\left(\wedge_{\ell=0}^{k} \mathcal{E}_\ell\right) \geq 1 - \sum_{\ell=0}^{k} \Pr(\neg \mathcal{E}_\ell) \geq 1/4 \tag{8}$$

Since u is Gaussian, λ, z can be viewed as independent gaussian random variables. So we have that $\mathcal{E}_{k+1}$ is independent of $\mathcal{E}_\ell, \ell = 0, 1, 2, \ldots, k$. Therefore, Fact 4.4 and (8) imply that

$$\Pr\left(\wedge_{\ell=0}^{k+1} \mathcal{E}_\ell\right) \geq \frac{1}{40} k^{-10/\delta^2}.$$

□

For rest of the proof, we assume that the event $\wedge_{\ell=0}^{k+1} \mathcal{E}_\ell$ occurs. Now, for all $\ell \in [k]$,

$$u \cdot (\zeta_\ell - b) \overset{(7)}{\leq} z \cdot (\zeta_\ell - b) \leq 2\sqrt{\ln k}\Delta(K), \tag{9}$$

where the last inequality follows from the fact that the event $\mathcal{E}_\ell$ has occured. Also, since $z \perp (a-b)$:

$$u \cdot (a-b) = \lambda |a-b| \geq 3\sqrt{\ln k}\Delta(K), \tag{10}$$

where the last inequality follows from the fact that the event $\mathcal{E}_{k+1}$ has occured and $|a-b| \geq \delta\Delta(K)$. The above two inequalities along with (6) imply that

$$u \cdot a - \max_{y \in K} u \cdot y \geq \lambda |a-b|/3.$$

Now, event $\mathcal{E}_0$ implies that

$$|u| \leq \lambda + |z| \leq \lambda + 4\sqrt{m},$$

and so,

$$u \cdot a - \max_{y \in K} u \cdot y \geq \frac{\lambda \delta \Delta(K)|u|}{3(\lambda + 4\sqrt{m})},$$

The desired result now follows from Corollary 4.5 and the fact that $\lambda \cdot \delta \geq 3\sqrt{\ln k}$ (Fact 4.4). □

5 From $\mathsf{OptOr}_\varepsilon(K)$ oracles to the Hausdorff Problem

We prove Theorem 1.9 in this section. We begin by defining Hausdorff distance formally.

Definition 5.1. The Hausdorff-distance, $\mathsf{Haus}(K, K')$, between two polytopes K and K' is the infimum over all values α such that the following condition is satisfied: for every point $x \in K$, there is a point $y \in K'$ such that $|x - y| \leq \alpha$, and vice versa.

We now give the formal version of Theorem 1.9:

Theorem 5.2. Suppose ε, δ are reals in $[0, 1]$ with

$$\delta > c\varepsilon\sqrt{d}, \delta > c/\sqrt{d}, \tag{11}$$

where c is a large enough constant. Let K be a k-vertex polytope with $\Delta(K)$ denoting the diameter of K. Suppose we are also given an $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{O}$. Let P be the set of answers from the oracle on $m := k^{10+c\delta^{-2}}$ independent random queries. Then, $\mathsf{Haus}(K, \mathsf{CH}(P)) \leq \delta \cdot \Delta(K)$.

Proof. We describe the algorithm for obtaining the desired set S (referred as Random Probes Algorithm in Section 1.3) in Algorithm 1. The set P is constructed as follows: we pick a set of m random unit vectors. For each such unit vector u, we add the corresponding point $x(u)$ returned by the oracle $\mathcal{O}$ to the set P.

Algorithm 1: Algorithm for finding the set P such that $\mathsf{CH}(P)$ approximates K.

1.1 Input: An $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{O}$.
1.2 Initialize a set P to $\varnothing$
1.3 Repeat m times:
1.4 Let u be a random unit vector in $\Re^d$.
1.5 Call $\mathcal{O}$ on u to get a vector $x(u)$.
1.6 Add $x(u)$ to P.
1.7 Output P.

One side of the desired result is easy to show:

Claim 5.3. For each $x \in \mathsf{CH}(P)$, there is a $y \in K$ such that $|x - y| \leq \delta\Delta(K)$.

Proof. For a point $x(u) \in P$, we know by the definition of $\mathsf{OptOr}_\varepsilon(K)$ oracle that there is a point $y \in K$ such that $|x(u) - y| \leq \varepsilon\Delta(K) \leq \delta\Delta(K)$, where $\varepsilon \leq \delta$ follows from (11). The desired result now follows from the convexity of K. □

It remains to show that for any vertex v of K, there is a point $y \in \mathsf{CH}(P)$ such that $|v - y| \leq \delta\Delta(K)$. Fix such a vertex v of K for rest of the discussion. Let the random unit vectors considered in Algorithm 1 (in the order they get generated) be $u^1, \ldots, u^m$. Let P^j denote the subset $\{x(u^1), \ldots, x(u^j)\}$ of P. Define an event $\mathcal{E}_j$ as follows:

$$\mathsf{dist}(v, \mathsf{CH}(P^j)) \leq \delta \cdot \Delta(K) \quad \text{or} \quad \mathsf{dist}(v, \mathsf{CH}(P^{j+1})) \leq \left(1 - \frac{\delta^2}{c'}\right) \mathsf{dist}(v, \mathsf{CH}(P^j)),$$

where c' is a large enough constant. Our main technical result is to show that conditioned on any choice of $u^1, \ldots, u^j$ the event $\mathcal{E}_j$ happens with reasonably high probability (where the probability is over the choice of u^{j+1}):

Lemma 5.4. For any index $j \in [m-1]$,

$$\Pr_{u^{j+1}}\left[\mathcal{E}_j | u^1, \ldots, u^j\right] \geq \frac{1}{100}.$$

Proof. Fix the vectors $u^1, \ldots, u^j$. If $\mathsf{dist}(v, \mathsf{CH}(P^j)) \leq \delta \cdot \Delta(K)$, then we are done. So assume this is not the case. Let b be the closest point in $\mathsf{CH}(P^j)$ to v. Thus,

$$|v - b| \geq \delta \cdot \Delta(K). \tag{12}$$

Define w as

$$w := \frac{v-b}{|v-b|}.$$

We can now express the vector u^{j+1} as $\lambda w + z$, where $\langle z, w \rangle = 0$. We first show the following useful properties of these vectors.

Claim 5.5. With probability at least $\frac{1}{100}$, the following three events happen:

$$|z| \leq 4\sqrt{d} \tag{13}$$

$$\max_{y \in K} |z \cdot (v - y)| \leq 2\sqrt{\ln k}\Delta(K) \tag{14}$$

$$\lambda \geq \frac{100}{\delta}\sqrt{\ln k} \tag{15}$$

Proof. The proofs of these three inequalities are identical to those of Fact 4.2, Fact 4.3 and Fact 4.4 (in order to prove (14), it suffices to show it for points y which are vertices of K). □

The following fact is also easy to show:

Fact 5.6.

$$|v - x(u^{j+1})| \leq 2\Delta(K).$$

Proof. By the definition of $\mathsf{OptOr}_\varepsilon(K)$ oracle, there is a point $p \in K$ such that $|p - x(u^{j+1})| \leq \delta \cdot \Delta(K) \leq \Delta(K)$. The desired result now follows by triangle inequality. □

Let δ_1 denote $\frac{\delta^2}{100}$. Let b_1 denote the vector

$$\delta_1 x(u^{j+1}) + (1 - \delta_1) b.$$

Since $b_1 \in \mathsf{CH}(P^{j+1})$, the desired result will follow if we prove the following:

$$|v - b_1|^2 \leq \left(1 - \frac{\delta^2}{100}\right)|v - b|^2. \tag{16}$$

Now,

$$\begin{aligned}
|v - b_1|^2 &= \delta_1^2 |v - x(u^{j+1})|^2 + (1-\delta_1)^2 |v-b|^2 + 2\delta_1(1-\delta_1)(v - x(u^{j+1})) \cdot (v - b) \\
&\overset{\text{Fact 5.6}}{\leq} 4\delta_1^2 \Delta(K)^2 + (1 - 2\delta_1)|v-b|^2 + \delta_1^2 \Delta(K)^2 + 2\delta_1(1-\delta_1)(v - x(u^{j+1})) \cdot (v-b) \\
&\overset{(12)}{\leq} \left(1 - \frac{3\delta_1}{2}\right)|v-b|^2 + +2\delta_1(1-\delta_1)|v-b| \cdot (v - x(u^{j+1})) \cdot w
\end{aligned}$$

$$= \left(1 - \frac{3\delta_1}{2}\right)|v-b|^2 + +2\delta_1(1-\delta_1)\left(\underbrace{\frac{|v-b|}{\lambda}\cdot(v - x(u^{j+1}))\cdot u^{j+1}}_{:=A}\right.$$

$$\left.\underbrace{-\frac{|v-b|}{\lambda}\cdot(v - x(u^{j+1}))\cdot z}_{:=B}\right) \tag{17}$$

We now bound each of the terms A and B above. Now,

$$A \leq \frac{|v-b|}{\lambda}\varepsilon\Delta(K) \overset{(11)}{\leq} \frac{|v-b|\delta\Delta(K)}{c\lambda} \overset{(12)}{\leq} \frac{|v-b|^2}{c\lambda} \overset{(15)}{\leq} \frac{|v-b|^2\delta}{c},$$

where the first inequality follows from the definition of $\mathsf{OptOr}_\varepsilon(K)$ oracle. We now bound the quantity B. Let y be the point in K closest to $x(u^{j+1})$. We know that $|y - x(u^{j+1})| \leq \varepsilon\Delta(K)$. Therefore,

$$\begin{aligned} B &\leq \frac{|v-b|}{\lambda}\left(|(v-y)\cdot z| + |z|\varepsilon\Delta(K)\right) \\ &\overset{(13),(14)}{\leq} \frac{|v-b|}{\lambda}\left(2\sqrt{\ln k}\Delta(K) + 4\varepsilon\sqrt{d}\Delta(K)\right) \\ &\overset{(11),(15)}{\leq} \frac{\delta|v-b|\Delta(K)}{10} \overset{(12)}{\leq} \frac{|v-b|^2}{10}. \end{aligned}$$

Substituting the above bound on A and B in (17) yields the desired result. □

We are now almost done. As the following result shows, it suffices to argue that enough number of events $\mathcal{E}_j$ happen:

Claim 5.7. If at least $\frac{c'}{\delta^2}\ln(2/\varepsilon)$ of the events $\mathcal{E}_j, j \in [m-1]$ happen, then $\mathsf{dist}(v, \mathsf{CH}(P)) \leq \delta\Delta(K)$.

Proof. Assume, for the sake of contradiction, that $\mathsf{dist}(v, \mathsf{CH}(P)) > \varepsilon\Delta(K)$. Assume that events $\mathcal{E}_{j_1}, \ldots, \mathcal{E}_{j_h}$ happen, where $h := \frac{c'}{\delta^2}\ln(2/\varepsilon)$. Now, for any index $i \in [h-1]$, the definition of $\mathcal{E}_{j_{i+1}}$ implies that

$$\mathsf{dist}(v, \mathsf{CH}(P^{j_{i+1}})) \leq \left(1 - \frac{\delta^2}{c'}\right)\mathsf{dist}(v, \mathsf{CH}(P^{j_{i+1}-1}) \leq \mathsf{dist}(v, \mathsf{CH}(P^{j_i})).$$

Therefore,

$$\mathsf{dist}(v, \mathsf{CH}(P^{j_h})) \leq \left(1 - \frac{\delta^2}{c'}\right)^h \mathsf{dist}(v, P^1) \overset{\text{Fact 5.6}}{\leq} \left(1 - \frac{\delta^2}{c'}\right)^h 2\Delta(K) \leq \delta\Delta(K).$$

□

It remains to show that with high probability at least $h := \frac{c'}{\delta^2}\ln(2/\varepsilon)$ of the events happen. In order to prove this, we divide the sequence $[m]$ into $[h]$ subsequences, each of length m/h. Call these subsequences $C_1, \ldots, C_h$. It follows from Lemma 5.4 that for any $i \in [h]$,

$$\Pr\left[\wedge_{j\in C_i}\neg\mathcal{E}_j\right] \leq 0.99^{m/h} \leq \frac{1}{h^2}.$$

A simple union bound now shows that with probability at least $1 - 1/h$, at least one event $\mathcal{E}_j$ happens during each of the subsequences $C_1, \ldots, C_h$. Claim 5.7 now proves the theorem.

□

6 From $\mathsf{OptOr}_\varepsilon(K)$ oracles to $\mathsf{ListLearn}$

We first define the notion of well-separatedness.

Definition 6.1. We say that a polytope K with vertex set V is δ-well-separated if for every vertex $v \in V$, we have

$$\mathsf{dist}(v, \mathsf{CH}(V \setminus \{v\})) \geq \delta \cdot \Delta(K),$$

where $\Delta(K)$ denotes the diameter of K.

We first show the lower bound result Theorem 1.11, which is restated here.

Theorem 1.11 (Oracle Lower Bound). The problem where, one is required to output a point which is within $\Delta(K)/10$ of some vertex of K, given only by an $\mathsf{OptOr}_\varepsilon(K)$ oracle, cannot be solved in deterministic polynomial time when $\varepsilon \geq 8 \ln d/\sqrt{d}$.

Proof. The proof is by producing an "adversarial oracle". The parameter ε is set to $8 \ln d/\sqrt{d}$. Our K will a "needle" of the form $\{\lambda u : \lambda \in [-1, 1]\}$, where, u is a unit length vector. The oracle's answer for each query will always be the 0 vector. Let $v_1, v_2, \ldots, v_q$, where $q = d^c$ for a constant c, be the vectors on which the oracle is queried. The answer 0 is clearly a valid answer for any $\mathsf{OptOr}_\varepsilon(K)$ oracle, provided u is a unit vector with $|u \cdot v_i| \leq 4 \ln d/\sqrt{d}$. For a single v_i, the probability that a random u satisfies $|u \cdot v_i| > 4 \ln d/\sqrt{d}$ is at most $e^{-d\varepsilon^2}$. So by union bound, we see that there is a vector u_1 satisfying

$$|u_1 \cdot v_i| \leq 4 \ln d/\sqrt{d} \forall i \in [q].$$

Further, by a similar argument, there is a vector $u_2$satisfying

$$|u_2 \cdot v_i| \leq 4 \ln d/\sqrt{d}, i \in [q] \; ; \; |u_1 - u_2|, |u_1 + u_2| \geq .1.$$

It follows that the two possible K, where

$$\{\lambda u_1 : \lambda \in [-1, 1]\} \; , \; \{\lambda u_2 : \lambda \in [-1, 1]\}$$

are both consistent with the answer 0 for all $v_i, i \in [q]$ and in addition, we have that no point $w \in \mathrm{R}^d$ is within distance ε of two vertices, one from each of these 2 needles. So no answer is valid for both needles and the adversarial oracle can choose one of the needles to render the algorithm's answer incorrect. □

We now formally prove Theorem 1.10, which gives an algorithm for the $\mathsf{ListLearn}$ problem.

Theorem 6.2. Suppose ε, δ are reals in $[0, 1]$ with $\delta^2 \geq c\varepsilon\sqrt{d}, \delta^3 \geq c\,\varepsilon$, where c is a large enough constant. Let K be a δ-well-separated k-vertex polytope. Suppose we are also given $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{O}$. Let W be the set of answers of the oracle to $m = poly(d) \cdot k^{\Omega(1/\delta^2)}$ independent random queries. Then for each vertex v of K, there is a point $v' \in W$ such that $|v - v'| \leq O(\delta^2 \Delta(K)/c)$.

Proof. The algorithm chooses a set U of $k^{\Omega(1/\delta^2)}$ unit length i.i.d. Gaussian vectors. For each $u \in U$, it calls the oracle $\mathcal{O}$ to find a vector $x(u)$. Let W denote the set $\{x(u) : u \in U\}$. We invoke Corollary 7.6 with the parameters:

$$\delta' := \delta/4, \; \varepsilon' := \frac{32\delta^2}{c}$$

on the set W. The algorithm in that result outputs a set Q. We output that Q as the approximation to the set of vertices of K. We now prove that the set Q has the desired properties. We first show that for every vertex of K, there is a direction u in U along which the projection of this vertex is higher than the projection of the remaining vertices by a large enough margin. Let $M_{\cdot,1}, \ldots, M_{\cdot,k}$ be the vertices of K.

Claim 6.3. With high probability, the following event happens: for each $\ell \in [k]$, there is a vector $u^{(\ell)} \in U$ such that for all $\ell' \in [k], \ell' \neq \ell$, we have

$$u^{(\ell)} \cdot M_{\cdot,\ell} > u^{(\ell)} \cdot M_{\cdot,\ell'} + \frac{c\,\varepsilon \cdot \Delta(K)}{8\delta^2} \tag{18}$$

Proof. Fix a vertex $M_{\cdot,\ell}$ of K. Let K' be the convex hull of $\{M_{\cdot,1}, \ldots, M_{\cdot,k}\} \setminus \{M_{\cdot,\ell}\}$. We invoke Theorem 4.1 on the polytope K' and the point $a := M_{\cdot,\ell}$. The definition of δ-well-separated implies that (4) is satisfied with $\rho = \delta$. Using $V = \Re^d$ in the statement of Theorem 4.1, we see that

$$\Pr_{u \in U}\left[u \cdot M_{\cdot,\ell} - \max_{\ell' \in [k], \ell' \neq \ell} u \cdot M_{\cdot,\ell'} \geq \frac{\delta\sqrt{\log k}\Delta(K)}{\sqrt{\log k} + 4\delta\sqrt{d}}\right] \geq \frac{1}{40}k^{-10/\delta^2}.$$

Since

$$\frac{\delta\sqrt{\log k}}{\sqrt{\log k} + 4\delta\sqrt{d}} \geq \min\left(\frac{\delta}{2}, \frac{1}{8\sqrt{d}}\right) \geq \frac{\varepsilon c}{8\delta^2},$$

the desired result follows from the fact that $|U| \gg \frac{1}{40}k^{-10/\delta^2}$.

□

For rest of the proof, assume that the statement in Claim 6.3 holds true, i.e., there are directions $u^{(1)}, \ldots, u^{(k)} \in U$ satisfying (18). We now show that for every vertex of $M_{\cdot,\ell}$ of K, the corresponding point $x(u^{\ell})$ is close to $M_{\cdot,\ell}$.

Claim 6.4. For every $\ell \in [k]$,

$$|x(u^{(\ell)}) - M_{\cdot,\ell}| \leq 17\delta^2\Delta(K)/c.$$

Proof. By the definition of $\mathcal{O}$, we know that $x(u^{(\ell)})$ can be written as $y(u^{\ell}) + z(u^{(\ell)})$, where $y(u^{(\ell)}) \in K$ and $|z(u^{(\ell)})| \leq \varepsilon\Delta(K)$. Thus, there is a convex combination $\lambda_{\ell'}, \ell' \in [k]$, of the vertices $M_{\cdot,\ell'}$ of K such that

$$x(u^{(\ell)}) = \sum_{\ell' \in [k]} \lambda_{\ell'} M_{\cdot,\ell'} + z(u^{(\ell)}).$$

By the definition of $\mathcal{O}$, $x(u^{(\ell)}) \cdot u^{(\ell)} \geq M_{\cdot,\ell} \cdot u^{(\ell)} - \varepsilon\Delta(K)$ and $|z(u^{(\ell)}) \cdot u^{(\ell)}| \leq \varepsilon\Delta(K)$. So, we get

$$M_{\cdot,\ell} \cdot u^{(\ell)} - \varepsilon\Delta(K) \leq \sum_{\ell' \in [k]} \lambda_{\ell'} M_{\cdot,\ell'} \cdot u^{(\ell')} + \varepsilon\Delta(K),$$

which implies (after subtracting $\lambda_\ell M_{\cdot,\ell} \cdot u^{(\ell)}$ from both sides):

$$(1 - \lambda_\ell) M_{\cdot,\ell} u^{(\ell)} \leq \sum_{\ell' \neq \ell} \lambda_{\ell'} M_{\cdot,\ell'} \cdot u^{(\ell')} + 2\varepsilon\Delta(K)$$

which, using Claim 6.3, yields:

$$(1 - \lambda_\ell) M_{\cdot,\ell} u^{(\ell)} \leq (1 - \lambda_\ell) M_{\cdot,\ell} u^{(\ell)} - (1 - \lambda_\ell)\frac{c\varepsilon\Delta(K)}{8\delta^2} + 2\varepsilon\Delta(K).$$

It follows from the above inequality that

$$1 - \lambda_\ell \leq \frac{16\delta^2}{c}. \tag{19}$$

Therefore,

$$|x(u^{(\ell)}) - M_{\cdot,\ell}| = \left|\sum_{\ell' \neq \ell} \lambda_{\ell'}(M_{\cdot,\ell} - M_{\cdot,\ell'})\right| + \varepsilon\Delta(K) \leq \sum_{\ell' \neq \ell} \lambda_{\ell'}\Delta(K) + \varepsilon\Delta(K) \overset{(19)}{\leq} \frac{17\delta^2\Delta(K)}{c}.$$

□

This completes proof of the Theorem. □

The statement of Theorem 6.2 requires two different bounds relating δ to ε. In some applications, it may be difficult to ensure both of these conditions. We consider the setting when $d \gg k$, where the following variation of Theorem 6.2 is better suited.

Theorem 6.5. Suppose ε, δ are reals in $[0,1]$ with $\delta^2 \geq c\varepsilon\sqrt{d}, \delta \geq \frac{\sqrt{\log k}}{\sqrt{cd}}$, where c is large enough constant. Let K be a δ-well-separated k-vertex polytope. Suppose we are also given an $\mathsf{OptOr}_{\varepsilon}(K)$ oracle $\mathcal{O}$. Let W be the set of answers of the oracle to $m = poly(d) \cdot k^{O(1/\delta^2)}$ independent queries. Then for each vertex v of K, there is a point $v' \in W$ such that $|v - v'| \leq O(\delta^2 \Delta(K)/\sqrt{c})$.

Proof. The proof is identical to that of Theorem 6.2, except that in the proof of Claim 6.3, we now have the following modified argument. Since $\delta\sqrt{cd} \geq \sqrt{\log k}$, we get

$$\frac{\delta\sqrt{\log k}}{\sqrt{\log k} + 4\delta\sqrt{d}} \geq \frac{1}{8\sqrt{cd}} \geq \frac{\varepsilon\sqrt{c}}{8\delta^2}.$$

Rest of the proof of Claim 6.3 follows without any further changes (where we replace c by $\sqrt{c}$). □

7 From ListLearn to the k-OLP Problem

In this section, we show that for well-separated polytopes, a solution for the ListLearn problem can be used to solve the k-OLP problem as well. This algorithm uses the notion of soft convex hulls. We first describe the algorithm for constructing soft convex hulls, and then use it to solve the k-OLP problem.

7.1 Soft Convex Hulls

Let W be a finite set of points in $\Re^d$, and T be the vertices of $\mathsf{CH}(W)$. The subset T of W is the unique subset of W with the following properties:

(P1) $W \subseteq \mathsf{CH}(T)$

(P2) $\forall w \in W$, if $w \notin \mathsf{CH}(W \setminus \{w\})$, then $w \in T$.

We now define a natural notion of soft convex hull.

Definition 7.1. For an $\varepsilon \geq 0$, and $S \subseteq W$, define the ε-convex hull of S, ε-$\mathsf{CH}(S)$, as $\mathsf{CH}(S) + \varepsilon\Delta(W)B$, where B is the unit ball of the Eucliean norm.

The intuition behind the above definition is that $\mathsf{CH}(W)$ can have a very large set of vertices, but there may be a small set of points whose soft convex hull contains W. This is defined more formally as follows:

Definition 7.2. We call a subset $T \subseteq W$ an ε-envelope of W, written ε-$\mathsf{ENV}(W)$, if $W \subseteq \varepsilon$-$\mathsf{CH}(T)$.

Remarks: The following observations about the set ε-$\mathsf{ENV}(W)$ are easy to see:

(a) There are several distinct sets T which could qualify as ε-$\mathsf{ENV}(W)$. For example, let W consist of the following set of points in $\Re^2$: a set of points W_1 close to $(0,0)$ and a set of points W_2 close to $(1,0)$. Let T be a pair of points $\{x, y\}$ with $x \in W_1, y \in W_2$. Then it is easy to check that T is an ε-$\mathsf{ENV}(W)$.

(b) Let T be ε-$\mathsf{ENV}(W)$. Unlike property (P2) above, it is not necessary that if $w \in W$ is such that $w \notin \varepsilon$-$\mathsf{CH}(W \setminus \{w\})$, then $w \in T$.

Since the set ε-$\mathsf{ENV}(W)$ is not uniquely determined, we will impose one more condition on it to make it unique (if it exists) and polynomial time computable. This condition requires the points of T to be "far apart" from each other. More precisely:

Definition 7.3. For $\varepsilon, \delta \in [0,1]$, a set T is called a (ε,δ)-$\mathsf{ENV}(W)$ if it is an ε-$\mathsf{ENV}(W)$ and

$$\forall w \in T, \mathsf{dist}(w, \mathsf{CH}(T \setminus \{w\})) > \delta \Delta(W) \tag{20}$$

Fact 7.4. Given a subset T of W, we can check in polynomial time whether T is an (ε,δ)-$\mathsf{ENV}(W)$.

Proof. Fix a subset T. We can verify in polynomial time whether T is an ε-$\mathsf{ENV}(W)$. Indeed, for each point $w \in W$, we need to check if $w \in \varepsilon$-$\mathsf{CH}(T)$. This can be expressed as a convex program feasibility problem, where the variables are λ_t for each $t \in T$:

$$\left| \sum_{t \in T} \lambda_t \cdot t - w \right| \leq \varepsilon \Delta(W), \quad \sum_t \lambda_t = 1, \quad \lambda_t \geq 0 \ \ \forall t \in T.$$

Similarly, we can check (20) using convex programming. For each $w \in T$, we can find $\mathsf{dist}(w, \mathsf{CH}(T \setminus \{w\})$ as follows, where the variables are $\lambda_x, x \in T \setminus \{w\}$:

$$\min . \left| \sum_{x \in T \setminus \{w\}} \lambda_x \cdot x - w \right|, \quad \sum_x \lambda_x = 1, \quad \lambda_x \geq 0 \ \ \forall x \in T \setminus \{w\}.$$

□

For rest of the section, we address the following question: given the set W, and parameters ε, δ, is there a (ε,δ)-$\mathsf{ENV}(W)$, and if so, can we find in polynomial time an approximation to this set? We informally argue that several "natural" greedy strategies do not work. Consider, for example, the following algorithms for identifying (ε,δ)-$\mathsf{ENV}(W)$:

- Identify T as the set of points $w \in W$ for which w is not close to $\mathsf{CH}(W \setminus \{w\})$: We define a set of points in $\Re^2$. We have rings of points, one around $(0,0)$ and the other around $(0,1)$, each point is close to the convex hull of the others and so this algorithm will dismiss all the points as not belonging to the desired set T.

- Start by adding an extreme point (along an arbitrarily chosen direction) to T. For the next $k-1$ steps, iteratively find a point w for which $\mathsf{dist}(w, \mathsf{CH}(T))$ is maximized and add it to T: a simple example as above shows that this idea may not work as well. Consider v_1, v_2, v_3, v_4 lying on corners of a square, and let v_5 be the mid-point of v_3, v_4. We start by adding v_1 and then v_2 to T. But in the next step, we could add v_5, and then both v_3, v_4. But the "correct" T would have been $\{v_1, v_2, v_3, v_4\}$.

If $\varepsilon = 0$, the above question is easy to answer in polynomial time. The answer is yes iff the set T of vertices of $\mathsf{CH}(W)$ satisfies (20). Also, if $\delta = 1$, T has to be a singleton to satisfy (20).

In rest of this section, we consider the following problenm: For what pairs of values of ε, δ, can we prove that there is essentially at most one (ε,δ)-$\mathsf{ENV}(W)$, and if so, can we determine this set efficiently ? We do not know the exact answer to this, but our main result here (which suffices for the applications) is (verbally stated) an affirmative answer to the question if the following condition is satisfied:

$$\delta \in \Omega(\sqrt{\varepsilon}).$$

This will follow as a corollary of our main result:

Theorem 7.5. Let $\delta, \varepsilon, \varepsilon_3$ be reals in $(0, 1/8)$ satisfying

$$\delta > \max\left(\frac{2\varepsilon}{\varepsilon_3 - \varepsilon}, 4\varepsilon_3\right) \tag{21}$$

Let W be a finite set of points in $\Re^d$. We can determine in polynomial time whether exists a set T in (ε, δ)-$\mathsf{ENV}(W)$, and if so, we can efficiently find a subset Q of W such that

$$|Q| = |T| \tag{22}$$

$$\forall w \in T, \exists x \in Q : |w - x| \leq 2\varepsilon_3 \Delta(W) \tag{23}$$

Corollary 7.6. Let δ, ε be reals in $(0, 1/8)$ satisfying $\delta > 16\sqrt{\varepsilon}$ Let W be a finite set of points in $\Re^d$. We can determine in polynomial time whether exists a set T forming a (ε, δ)-$\mathsf{ENV}(W)$, and if so, we can efficiently find a subset Q of W such that

$$|Q| = |T| \tag{24}$$

$$\forall w \in T, \exists x \in Q : |w - x| \leq 8\sqrt{\varepsilon}\Delta(W) \tag{25}$$

Proof. The Corollary follows from Theorem (7.5) by taking $\varepsilon_3 = 4\sqrt{\varepsilon}$ □

Proof. (of Theorem 7.5) The procedure is described in Algorithm 2. We first compute a subset Q'' of W consisting of points $w \in W$ which do not lie in the soft convex hull of the points in W which are "far" from w – this can be doen in polynomial time by using arguments similar to those in the proof of Fact 7.4. Then Q is defined as a maxumal subset of points in Q'' with the property that the pair-wise distance between the points in it large. Finally, we check whether Q is an (ε, δ)-$\mathsf{ENV}(W)$, which can be done efficiently using Fact 7.4.

Algorithm 2: Procedure to identify (ε, δ)-$\mathsf{ENV}(W)$

2.1 Compute $\Delta(W)$.
2.2 Let $Q' := \{w \in W : w \in \varepsilon\text{-}\mathsf{CH}\left(\{x \in W : |w - x| \geq \varepsilon_3 \Delta(W)\}\right).$
2.3 Let $Q'' := W \setminus Q'$.
2.4 Define $Q :=$ maximal subset of Q'' such that for every distinct $x, y \in Q$, $|x - y| > 2\varepsilon_3\Delta(W)$.
2.5 if Q is (ε, δ)-$\mathsf{ENV}(W)$ then
2.6 ⌊ Output the set Q.
2.7 else
2.8 ⌊ Output No.

Now we analyze this algorithm. If there is no (ε, δ)-$\mathsf{ENV}(W)$, then the algorithm will clearly say "No" (because the set Q will not be (ε, δ)-$\mathsf{ENV}(W)$). So assume that there is a set T which is (ε, δ)-$\mathsf{ENV}(W)$.

Claim 7.7. $T \subseteq Q''$.

Proof. Suppose for the sake of contradiction that there is a point $w \in T \setminus Q''$. For sake of brevity, let W_x denote the subset of points in W satisfying $|w - x| \geq \varepsilon_3 \Delta(W)$. The fact that $w \notin Q''$ implies that $w \in \mathsf{CH}(W_x)$, i.e.,

$$w = \sum_{x \in W_x} \lambda_x \cdot x + e_0,$$

where λ_x form a convex combination and $|e_0| \leq \varepsilon\Delta(W)$. Since $W \subseteq \varepsilon$-$\mathsf{ENV}(T)$, for each $x \in W_x$, there are a point $x' \in \mathsf{CH}(T)$ such that $e_x := x - x'$ has length at most $\varepsilon\Delta(W)$. Since $w \in T$, we can write x' as

$$x' = \mu_x w + (1 - \mu_x) y_x, \quad y_x \in \mathsf{CH}(T \setminus \{w\}),$$

where $\mu_x \in [0,1]$. Therefore,

$$|w-x| \le |w-x'| + |e_x| = (1-\mu_x)|w-y_x| + |e_x| \le (1-\mu_x)\Delta(W) + \varepsilon\Delta(W).$$

But we know that $|w-x| \ge \varepsilon_3\Delta(W)$. Therefore, we get

$$(1-\mu_x) \ge \varepsilon - \varepsilon_3. \tag{26}$$

Now,

$$\begin{aligned} w &= \sum_{x\in W_x} \lambda_x \cdot x + e_0 = \sum_{x\in W_x} \lambda_x \cdot x' + \sum_x \lambda_x e_x + e_0 \\ &= \sum_{x\in W_x} \lambda_x\mu_x \cdot w + \sum_{x\in W_x} \lambda_x(1-\mu_x)y_x + e, \end{aligned}$$

where $e = \sum_x \lambda_x e_x + e_0$ has length at most $2\varepsilon\Delta(W)$. Let θ_x denote $\lambda_x(1-\mu_x)$. Observe that $\theta := \sum_x \theta_x = 1 - \sum_x \lambda_x\mu_x$. Therefore, we can rewrite the above as

$$w - \frac{1}{\theta}\sum_{x\in W_x} \theta_x y_x = \frac{e}{\theta}.$$

Since $\frac{1}{\theta}\sum_{x\in W_x} \theta_x y_x \in \mathsf{CH}(T\setminus\{w\})$, we see that

$$\mathsf{dist}(w, \mathsf{CH}(T\setminus\{w\})) \le \frac{|e|}{\theta} \le \frac{2\varepsilon\Delta(W)}{\sum_{x\in W_x}\lambda_x(1-\mu_x)} \overset{(26)}{\le} \frac{2\varepsilon}{\varepsilon-\varepsilon_3}\Delta(W).$$

Using (20) and (21), we get a contradiction. Therefore, $w \in Q''$. □

Since $T \subseteq Q''$, for every $w \in T$, there exists a point $x_w \in Q$ such that $|w - x_w| \le 2\varepsilon_3\Delta(W)$. It is easy to see that if w, w' are two distinct points in T, then $x_w \ne x_{w'}$. Indeed,

$$|x_w - x_{w'}| \ge |w-w'| - |w-x_w| - |w'-x_{w'}| \ge \delta\Delta(W) - 4\varepsilon_3\Delta(W) \overset{(21)}{>} 0.$$

Finally, we prove that every element in Q is of the form x_w for some $w \in T$.

Claim 7.8. Every $x \in Q$ is of the form x_w for some $w \in T$.

Proof. Consider a point $x \in Q$. This implies that $x \in Q''$. We claim that there is a point $w \in T$ such that $|x-w| \le \varepsilon_3\Delta(W)$. Suppose not. Then the set of points w such that $|x-w| \ge \varepsilon_3\Delta(W)$ includes T. But we know that $x \in \varepsilon\text{-}\mathsf{CH}(T)$, and so $x \in Q'$, a contradiction.

Therefore

$$|x - w_x| \le |x-w| + |w-w_x| \le 2\varepsilon\Delta(W).$$

This implies that $x = w_x$ (since any two distinct points in Q have distance greater than $2\varepsilon\Delta(W)$. □

Thus we have shown that $|Q| = |T|$ and for every $w \in T$, there is a unique element $x_w \in Q$ with $|w - x_w| \le 2\varepsilon_3\Delta(W)$. □

7.2 Algorithm for k-OLP

We now show how soft convex hulls can be used to generate a solution for the k-OLP problem. The following result, which formalizes Theorem 1.13, uses the same setting as that in Theorem 6.2:

Theorem 7.9. Suppose ε, δ are reals in $[0,1]$ with $\delta^2 \geq c\varepsilon\sqrt{d}, \delta^3 \geq c\,\varepsilon$, where c is a large enough constant. Let K be a δ-well-separated k-vertex polytope. Suppose we are also given an $\mathsf{OptOr}_\varepsilon(K)$ oracle $\mathcal{O}$. Let W be the set of answers of the oracle $\mathcal{O}$ to $m = poly(d) \cdot k^{\Omega(1/\delta^2)}$ independent random queries. We can find $Q \subseteq W, |Q| = k$ in randomized $poly(d) \cdot k^{\Omega(1/\delta^2)}$-time which satisfies the following condition w.h.p.: for every vertex v of K, there is a point v' in Q with $|v - v'| \leq \delta\Delta(K)/10$.

Proof. The proof of Theorem 6.2 shows that for every vertex $M_{\cdot,\ell}$ of K, there is point $x(u^\ell) \in W$ such that

$$|x(u^{(\ell)}) - M_{\cdot,\ell}| \leq 17\delta^2\Delta(K)/c. \tag{27}$$

Let T denote the set of points $\{x(u^{(\ell)}) : \ell \in [k]\}$. Our first claim is that the points of T are also well-separated.

Claim 7.10. For any $\ell \in [k]$,

$$\mathsf{dist}(x(u^{(\ell)}), \mathsf{CH}(T \setminus \{x(u^{(\ell)})\})) \geq \delta\Delta(K)/2.$$

Further, the diameter of $\mathsf{CH}(T)$ is at most $2\Delta(K)$.

Proof. Fix an index $\ell \in [k]$ and a point $y \in \mathsf{CH}(T \setminus \{x(u^{(\ell)})\})$. We can express y as a convex combination of points in $T \setminus \{x(u^{(\ell)})\}$, i.e.,

$$y = \sum_{\ell' \neq \ell} \lambda_{\ell'} \cdot x(u^{(\ell')}), \quad \text{where} \quad \sum_{\ell' \neq \ell} \lambda_{\ell'} = 1.$$

Now

$$\begin{aligned} |x(u^{(\ell)}) - y| &\geq \left| M_{\cdot,\ell} - \sum_{\ell' \neq \ell} \lambda_{\ell'} \cdot M_{\cdot,\ell'} \right| - |x(u^{(\ell)}) - M_{\cdot,\ell}| - \sum_{\ell' \neq \ell} \lambda_{\ell'} \cdot |x(u^{(\ell')}) - M_{\cdot,\ell'}| \\ &\geq \left(\delta - \frac{34\delta^2}{c}\right) \Delta(K) \geq \delta\Delta(K)/2, \end{aligned}$$

where the second last inequality follows from (27) and the fact that K is δ-well-separated. Since $\mathsf{dist}(x(u^{(\ell)}), K) \leq \varepsilon\Delta(K)$, it follows that $\Delta(\mathsf{CH}(T)) \leq 2\Delta(K)$. □

Recall that W denotes the set $\{x(u) : u \in U\}$. We now show that $\mathsf{CH}(T)$ closely approximates the set $\mathsf{CH}(W)$.

Claim 7.11. $W \subseteq \varepsilon'\text{-}\mathsf{CH}(T)$, where $\varepsilon' = \frac{32\delta^2}{c}$.

Proof. Fix a point $x(u) \in W$. We know that $x(u)$ can be written as

$$x(u) = y(u) + z(u), \quad y(u) \in K, |z(u)| \leq \varepsilon.$$

Let $y(u) = \sum_{\ell \in [k]} \lambda_\ell \cdot M_{\cdot,\ell}$, where the coefficients λ_ℓ form a convex combination. Then

$$\left| x(u) - \sum_{\ell \in [k]} \lambda_\ell \cdot x(u^{(\ell)}) \right| \leq \sum_{\ell \in [k]} \lambda_\ell \cdot |x(u^{(\ell)}) - M_{\cdot,\ell}| + |z(u)| \leq \frac{17\delta^2\Delta(K)}{c} + \varepsilon\Delta(K) \leq \frac{32\delta^2\Delta(K)}{c},$$

where the second last inequality follows from (27) and the last inequality by the assumption in. This proves the desired result. □

Claim 7.10 and Claim 7.11 imply that T is (ε', δ')-ENV(W) with $\delta' = \delta/4, \varepsilon' = \frac{32\delta^2}{c}$. We can now apply Corollary 7.6 to get approximations to $x(u^{(\ell)})$ within distance $17\sqrt{\varepsilon'}\Delta(K)$. Claim 7.11 now implies that we can get approximations to $M_{\cdot,\ell}$ within distance

$$17\sqrt{\varepsilon'}\Delta(K) + \frac{16\delta^2\Delta(K)}{c} \leq \frac{\delta\Delta(K)}{10}.$$

This proves the desired result. □

The following result is the analogue of Theorem 6.5, and gives an algorithm for the k-OLP problem under slightly different conditions on the parameters δ and ε.

Theorem 7.12. Suppose ε, δ are reals in $[0, 1]$ with $\delta^2 \geq c\varepsilon\sqrt{d}, \delta \geq \frac{\sqrt{\log k}}{\sqrt{cd}}$, where c is large enough constant. Let K be a δ-well-separated k-vertex polytope. Suppose we are also given an OptOr$_\varepsilon(K)$ oracle $\mathcal{O}$. Let W be the set of answers of the oracle $\mathcal{O}$ to $m = poly(d) \cdot k^{O(1/\delta^2)}$ independent queries. In randomized $poly(d) \cdot k^{O(1/\delta^2)}$-time we can find $Q \subset W$ of k points such that the following condition is satisfied w.h.p.: for every vertex v of K, there is a point v' in Q with $|v - v'| \leq \delta\Delta(K)/10$.

8 k-OLP algorithm for Latent Polytopes using Singular Value Decomposition

Theorem 1.11 showed that a solution to k-OLP problem requires the error parameter ε to be $O^*(1/\sqrt{d})$. Theorem 6.2 and Theorem 6.5 give algorithms achieving this bound. However, for many polytopes with k vertices, we can solve the k-OLP problem with ε being $O^*(1/\sqrt{k})$. However, if $k < d$, this error is too high. To tackle this, we find a good approximation to the subspace spanned by the vertices of K, then we project to this subspace and use the result in Theorem 6.2. One such example is the "Latent $k-$ Polytope" (abbreviated LkP) problem which we now describe.

The LkP problem has been studied in [BK20]. Certain assumptions were made on the model, namely, the hidden polytope K as well as on the (hidden) process for generating observed data from latent points in K. These assumptions are (a) shown to hold in several important Latent Variable models and (b) are sufficient to enable one to get polynomial time learning algorithms.

Here, we formulate assumptions which are similar, but, weaker in one important aspect. Whereas [BK20] assumed that each vertex of K has a separation from the affine hull of the other vertices (thus, in particular, each vertex is affinely independent of other vertices), we assume here that each vertex is separated only from the convex hull of the others. Under this weaker assumption, the algorithm of [BK20] does not work. We give a different algorithm which we prove works. It is also simpler to state and carry out and its proof is based on a new general tool we introduce here - the Random Separating Hyperplane theorem (Theorem 4.1).

Assumptions on data in the LkP problem: Let $M_{\cdot,1}, \ldots, M_{\cdot,k}$ denote the vertices of K and M be the $d \times k$ matrix with columns representing the vertices of K. We assume there are latent (hidden) points $P_{\cdot,j}, j = 1, 2, \ldots, n$ in K and observed data points $A_{\cdot,j}, j = 1, 2, \ldots, n$ are generated (not necessarily under any stochastic assumptions) by adding displacements $A_{\cdot,j} - P_{\cdot,j}$ respectively to $P_{\cdot,j}$. Clearly if the displacements are arbitrary, it is not possible to learn K given only the observed data. So we need some bound on the displacements.

Secondly, if all (or almost all) latent points lie in (or close to) the convex hull of a subset of $k-1$ or fewer vertices of K, the missing vertex cannot be learnt. To avoid this, we will assume that there is a certain w_0 fraction of latent points close to every vertex of K.

Let [4]

$$\sigma_0 := \frac{||\mathrm{P} - \mathrm{A}||}{\sqrt{n}}. \tag{28}$$

[4]By the standard definition of spectral norm, it is easy to see that σ_0^2 is the maximum mean squared displacement in any direction.

We now show that the k-OLP-Algorithm mentioned in Section 1.3 has the desired properties:

Theorem 8.1. Suppose K is a latent poytope with k vertices $M_{\cdot,1}, M_{\cdot,2}, \ldots, M_{\cdot,k}$ and P, A are latent points (all in K) and observed data respectively. Assume

$$\text{For all } \ell \in [k]\ ,\ C_\ell := \{j : |P_{\cdot,j} - M_{\cdot,\ell}| \leq \frac{\sigma_0}{\sqrt{w_0}}\} \text{ satisfies } |C_\ell| \geq w_0 n. \tag{29}$$

Suppose $(\sqrt{\log k}/\sqrt{c_0 k}) \leq \delta \leq 1$ and c_0 is a large constant satisfying

$$\sigma_0 \leq \frac{\delta^2 \Delta(K)}{100 c_0} \frac{\sqrt{w_0}}{\sqrt{k}}. \tag{30}$$

Let V be the k-dimensional SVD subspace of A, and $\widehat{K}$ denote the projection of K on V.

- There is an $\mathsf{OptOr}_{\frac{10\sigma_0}{\sqrt{w_0}\Delta}}(\widehat{K})$ oracle $\mathcal{O}$.
- The algorithm k-OLP Algorithm in Section 1.3 outputs a set Q of k points such that the following condition is satisfied w.h.p.: for every vertex v of K, there is a point v' in Q with $|v - v'| \leq \delta\Delta(K)/5$.

In the Theorem above, (30) implies an upper bound on σ_0 of $\Delta(K)\sqrt{w_0}/(c_0\sqrt{k})$ and so the oracle $\mathcal{O}$ used by the Theorem lies in $\mathsf{opt}_\varepsilon(\widehat{K})$ for $\varepsilon \leq 1/(c_0\sqrt{k})$. Thus, we get around the lower bound result Theorem 1.11 by working in the k-dimensional SVD subspace of A. In rest of this section, we prove Theorem 8.1. We begin by giving properties of the SVD subspace, and then show that an approximate optimization oracle exists in this subspace. Finally, we apply Theorem 7.12.

8.1 Properties of the SVD subspace of A

Let V denote the k-dimensional SVD subspace of A. We shall show that the projection of K on V has an $\mathsf{OptOr}_\varepsilon(K)$ oracle for a suitable value of ε. Let $\widehat{M}_{\cdot,\ell}$ denote the projection of $M_{\cdot,\ell}$ on the SVD subspace V. Define $\widehat{A}_{\cdot,j}$ similarly, and let $\widehat{\mathrm{A}}$ be the $d \times n$ matrix whose columns are given by $\widehat{A}_{\cdot,j}$. We now show that $\widehat{M}_{\cdot,\ell}$ and $M_{\cdot,\ell}$ are close to each other. The following notation will turn out to be very useful in subsequent discussion: for a $d \times n$ matrix B and a subset S of $[n]$, define

$$B_{\cdot,S} := \frac{1}{|S|} \sum_{j \in S} B_{\cdot,j}.$$

Claim 8.2. Let $S \subseteq [n]$ be a subset of indices. Then,

$$|A_{\cdot,S} - P_{\cdot,S}| \leq \frac{\sigma_0 \sqrt{n}}{\sqrt{|S|}}.$$

Proof. Define a unit vector $v \in \Re^d$ as follows: if $j \in S$, we define $v_j = \frac{1}{\sqrt{|S|}}$, 0 otherwise. Now, observe that

$$|A_{\cdot,S} - P_{\cdot,S}| = \frac{1}{\sqrt{|S|}} \cdot |(\mathrm{A} - \mathrm{P}) \cdot v|.$$

Inequality (28) now implies that the RHS above is at most $\frac{\sigma_0\sqrt{n}}{\sqrt{|S|}}$. □

Lemma 8.3. For all $\ell \in [k]$, $|M_{\cdot,\ell} - \widehat{M}_{\cdot,\ell}| \leq 5\sigma_0/\sqrt{w_0} \leq \frac{\delta^2\Delta(K)}{c_0}$.

Proof. We have $||\widehat{\mathrm{A}} - \mathrm{A}|| \leq ||\mathrm{A} - \mathrm{P}||$ since, $\widehat{\mathrm{A}}$ is the best rank k approximation to A in terms of the spectral norm and since each column of P being a convex combination of the columns $\{M_{\cdot,\cdot,\ell}, \ell \in [k]\}$, P has rank at most k. We also have $||\widehat{\mathrm{A}} - \widehat{\mathrm{P}}|| \leq ||\mathrm{A} - \mathrm{P}||$ since projections cannot increase length. Using these inequalities and the triangle inequality, we get:

$$||\mathrm{P} - \widehat{\mathrm{P}}|| \leq ||\mathrm{P} - \mathrm{A}|| + ||\mathrm{A} - \widehat{\mathrm{A}}|| + ||\widehat{\mathrm{A}} - \widehat{\mathrm{P}}|| \leq 3 \cdot ||\mathrm{A} - \mathrm{P}|| \overset{(28)}{\leq} 3\sigma_0\sqrt{n}.$$

Let C_ℓ be as in (29). Now, define w to be the unit vector with $w_j = 1/\sqrt{|C_\ell|}, \forall j \in C_\ell$, and $w_j = 0, \forall j \notin C_\ell$. We see that

$$|M_{\cdot,\ell} - \widehat{M}_{\cdot,\ell}| \leq \frac{|(\mathrm{P} - \widehat{\mathrm{P}})w|}{|C_\ell|} + \frac{2\sigma_0}{\sqrt{w_0}} \leq \frac{||\mathrm{P} - \widehat{\mathrm{P}}||}{\sqrt{|C_\ell|}} + \frac{2\sigma_0}{\sqrt{w_0}},$$

which proves the Lemma (using $|C_\ell| \geq w_0 n$). The second inequality in the Lemma follows from (30). □

Let $\widehat{K}$ be the projection of K on V. We now show that $\widehat{K}$ is also δ'-well-separated for a value δ' close to δ.

Lemma 8.4. The vertices of $\widehat{K}$ are given by $\{\widehat{M}_{\cdot,\ell} : \ell \in [k]\}$. Further, $\widehat{K}$ is δ'-well-separated, where $\delta' = \delta\left(1 - \frac{1}{100}\right)$.

Proof. It is clear that the vertex set of $\widehat{K}$ is s subset of $S := \{\widehat{M}_{\cdot,\ell} : \ell \in [k]\}$. Thus, we need to show that none of the points in S can be written as a convex combination of rest of the points in S. This will follow from the fact that $\widehat{K}$ is δ'-well-separated, and so it suffices to prove this statement. Consider a point $\widehat{M}_{\cdot,\ell} \in S$ and a point $p := \sum_{\ell' \in [k], \ell' \neq \ell} \lambda_{\ell'} \widehat{M}_{\cdot,\ell'}$, which is in $\mathsf{CH}(S \setminus \{\widehat{M}_{\cdot,\ell}\})$. Then

$$\begin{aligned}
|p - \widehat{M}_{\cdot,\ell}| &= \left|\sum_{\ell' \neq \ell} \lambda_{\ell'}(\widehat{M}_{\cdot,\ell'} - \widehat{M}_{\cdot,\ell})\right| \\
&\geq \left|\sum_{\ell' \neq \ell} \lambda_{\ell'}(M_{\cdot,\ell'} - M_{\cdot,\ell})\right| - \sum_{\ell' \neq \ell} \lambda_{\ell'}|M_{\cdot,\ell'} - \widehat{M}_{\cdot,\ell'}| - \sum_{\ell' \neq \ell} \lambda_{\ell'}|M_{\cdot,\ell} - \widehat{M}_{\cdot,\ell}| \\
&= \left|\sum_{\ell' \neq \ell} \lambda_{\ell'} M_{\cdot,\ell'} - M_{\cdot,\ell}\right| - \frac{2\delta^2\Delta(K)}{c_0} \\
&\geq (\delta - \frac{2\delta^2}{c_0})\Delta(K) \geq \delta'\Delta(K)
\end{aligned}$$

where the second last line uses Lemma 8.3 and last line follows from the fact that K is δ-well-separated. Note that $\Delta(\widehat{K}) \leq \Delta(K)$. Thus it follows that $\widehat{K}$ is is δ'-well-separated. □

We would now like to use Theorem 6.5 on $\widehat{K}$ with $\varepsilon = \frac{\sigma_0}{\sqrt{w_0}\Delta(K)}$ and $\delta' = \delta$. Indeed the condition (30) along with Lemma 8.4 show that the conditions of Theorem 6.5 hold provided we are able to exhibit an efficient $\mathsf{OptOr}_\varepsilon(\widehat{K})$ oracle.

8.2 Construction of the $\mathsf{OptOr}_\varepsilon(\widehat{K})$ oracle

We now describe the construction for the $\mathsf{OptOr}_\varepsilon(\widehat{K})$ oracle in the subspace V, where $\varepsilon = \frac{10\sigma_0}{\Delta(K)\sqrt{w_0}}$. Let u be a unit vector in V. The procedure (referred as Subset Smoothing Algorithm in Section 1.3) is given in Algorithm 3 – we project the points in $\widehat{A}$ on u and consider the $w_0 n$ with the highest projection along u. Finally, we output the average of these points.

Algorithm 3: Orcale in $\mathsf{OptOr}_{\varepsilon}(\widehat{K})$

3.1 Input: unit vector $u \in V$.

3.2 Let $R(u)$ be the index set of the columns $\widehat{A}_{\cdot,j}$ with the $w_0 n$ highest values of $u \cdot \widehat{A}_{\cdot,j}$.

3.3 Output: $\widehat{A}_{\cdot,R(u)}$

We now show that this algorithm has the desired properties. Let $R(u)$ be the index set as in Algorithm 3. The following is an easy consequence of Claim 8.2.

Claim 8.5. $\mathsf{dist}(\widehat{A}_{\cdot,R(u)}, \widehat{K}) \leq \varepsilon\Delta(K)$.

Proof. Claim 8.2 shows that $|A_{\cdot,R(u)} - P_{\cdot,R(u)}| \leq \varepsilon\Delta(K)$. Therefore, $|\widehat{A}_{\cdot,R(u)} - \widehat{P}_{\cdot,R(u)}| \leq \varepsilon\Delta(K)$. The desired result now follows because $\widehat{P}_{\cdot,R(u)} \in \widehat{K}$. □

Let $\ell \in [k]$ be the index for which $\widehat{M}_{\cdot,\ell} \cdot u$ is maximized.

Claim 8.6. $\widehat{A}_{\cdot,R(u)} \cdot u \geq \widehat{M}_{\cdot,\ell} \cdot u - \varepsilon\Delta(K)$.

Proof. Let C_ℓ be the index set specified by (29). It suffices to show that

$$\widehat{A}_{\cdot,C_\ell} \cdot u \geq \widehat{M}_{\cdot,\ell} \cdot u - \varepsilon\Delta(K).$$

Now,

$$\begin{aligned}
\widehat{M}_{\cdot,\ell} \cdot u - \widehat{A}_{\cdot,C_\ell} \cdot u &\leq |\widehat{M}_{\cdot,\ell} - \widehat{A}_{\cdot,C_\ell}| \\
&\leq |\widehat{M}_{\cdot,\ell} - \widehat{P}_{\cdot,C_\ell}| + |\widehat{P}_{\cdot,C_\ell} - \widehat{A}_{\cdot,C_\ell}| \\
&\leq |M_{\cdot,\ell} - P_{\cdot,C_\ell}| + |P_{\cdot,C_\ell} - A_{\cdot,C_\ell}| \\
&\leq \varepsilon\Delta(K),
\end{aligned}$$

where the last inequality follows from (29) and Claim 8.2. □

The above two results show that Algorithm 3 yields an $\mathsf{OptOr}_{\varepsilon}(\widehat{K})$ oracle. We would now like to apply Theorem 7.12 to the polytope $\widehat{K}$ with parameters ε and $\delta' = \delta(1 - 1/100)$. Lemma 8.4 shows that $\widehat{K}$ is δ'-well-separated. We now need to check that the parameters ε and δ' satisfy the following conditions needed in the statement of Theorem 7.12 (recall that $\widehat{K}$ is a polytope in $\Re^k$): (i) $\delta^2 \geq c\varepsilon\sqrt{k}$, and (ii) $\delta \geq \frac{\sqrt{\log k}}{\sqrt{ck}}$. The second condition is already an assumption in the statement of Theorem 8.1, and the first condition follows from (30) and the fact that $\varepsilon = \frac{10\sigma_0}{\Delta(K)\sqrt{w_0}}$. Applying Theorem 7.12, we get a set of k points Q, such that for each vertex v'' of $\widehat{K}$, there is a point $v' \in Q$ with $|v'' - v'| \leq \delta\Delta(K')/10$. Applying Lemma 8.3 proves Theorem 8.1.

9 Open Problems

We now mention some problems that remain open in our work:

(i) In the statement of RSH, the success probability of the desired event is $O\left(1/k^{O(1/\delta^2)}\right)$. Can we improve the exponential dependence of the success probability on $1/\delta$?

(ii) Prove an analog (under suitable assumptions) of Theorem 1.5 for other reductions among oracles. Of particular interest is a reduction from an Optimization oracle to a Separation Oracle.

(iii) Theorem 5.2 on the Haus problem returns exponentially many points whose convex hull approximates K. Can this be improved, either via a an improvement mentioned in the first open problem above, or alternatively, feeding the exponentially many points to the algorithm of [BHR19]?

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Appendix

A. Some Examples

The first example shows that the factor $1/\sqrt{m}$ in the statement of RSH (Theorem 1.1) cannot be improved, even for a simple polytope K consisting of just one line segment.

Example 1: Suppose K is a polytope in $\Re^d$ with just two vertices Δ distance apart and $V = \Re^d, \delta = 1$. It is easy to verify from standard results on random projection on a line that the additive term in inequality (2) is tight up to constant factors.

The next example shows that the success probability of the desired event (2) in Theorem 1.1 needs to depend on k, the number of vertices of the polytope K.

Example 2: Consider the Euclidean space $\Re^d$ and the $d-1$ dimensional sphere $K := \{x \in \Re^d : |x| \leq 1, x_1 = 0\}$. Observe that the diameter Δ of K is 1. Let a denote the point $(1, 0, 0, \ldots, 0)$. It is at distance 1 from K, and so the parameter δ in the statement of Theorem 1.1 can be set to 1. But, with high probability, a random unit length vector u has $u \cdot a \approx \frac{c}{\sqrt{d}}$ for some constant c, whereas the maximum of $v \cdot x$ over all points x in the sphere K is about 1. Therefore the probability of the event (2) is exponentially small in d.